for $m=n$

$$\int_{-\pi}^{\pi} \sin^2 nt \, dt$$

$$= \int_{-\pi}^{\pi} \frac{1-\cos 2nt}{2} \, dt$$

$$= \frac{1}{2}t\Big|_{-\pi}^{\pi} - \frac{1}{2}\frac{\sin 2nt}{2n}\Big|_{-\pi}^{\pi}$$

$$= \frac{1}{2}(\pi+\pi) - \frac{1}{4n}(0-0)$$

$$= \pi$$

$$\int_{-\pi}^{\pi} \sin^2 nt \, dt$$

$$= \int_{-\pi}^{\pi} \frac{1-\cos 2nt}{2} \, dt$$

$$= \frac{1}{2}\int_{-\pi}^{\pi} dt - \frac{1}{2}\int_{-\pi}^{\pi} \cos 2nt \, dt$$

$$= \frac{1}{2}(\pi+\pi) - 0$$

$$= \pi$$

For $m=n$

$$\int_{-\pi}^{\pi} \sin nt \cdot \cos nt \, dt$$

$$= \int_{-\pi}^{\pi} \frac{\sin 2nt}{2} \, dt$$

$$= \frac{1}{2}\frac{-\cos 2nt}{2n}\Big|_{-\pi}^{\pi}$$

$$= \frac{-1}{4n}(1-1) = 0$$

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